



Interactive Animations and Games

Lesson 8

Sprites

Warm Up





Review:

So far we've written programs that put simple shapes on the screen.

List of all of the different pieces of information that you have used to control how these shapes are drawn.

Journal Prompt:

If you wanted to create programs with more detailed images, maybe even characters that you could interact with, what other pieces of information might you need in your code?

Question of the Day

How can we use sprites to help us keep track of lots of information in our programs?

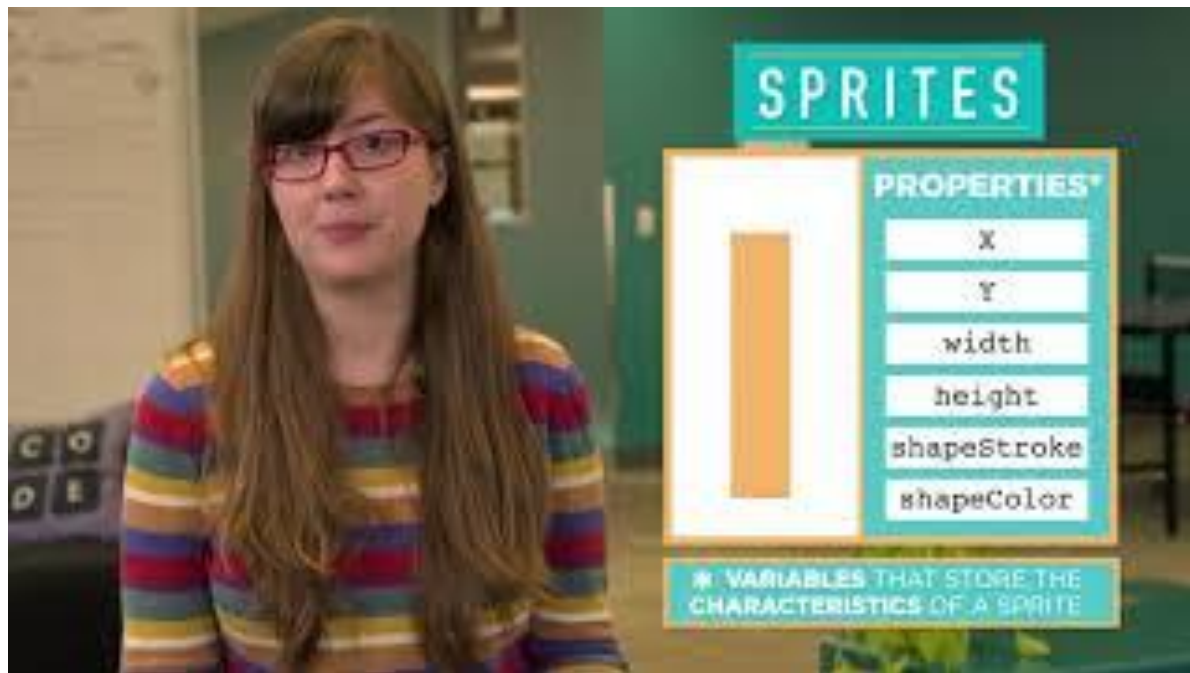
Activity



Code Studio, Level 1

Code Prediction: Sprites

- Read the code and discuss with a partner what will happen on the screen
- Write your predictions in the box, then press the Run button to see what happens
- Were you right?



What is a sprite?
What problem do sprites solve?



Key Vocabulary

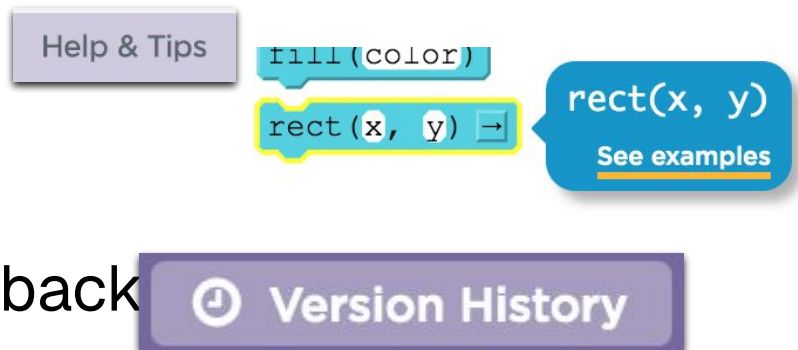
- **Sprite** - A character on the screen with properties that describe its location, movement, and look.

Code Studio, Level 2-3

Skill Building

Do This

- Follow the instructions on each level
- If you get stuck, try...
 - Help and Tips Tab
 - Code Documentation
- If you make a big mistake...
 - Use version history to go back





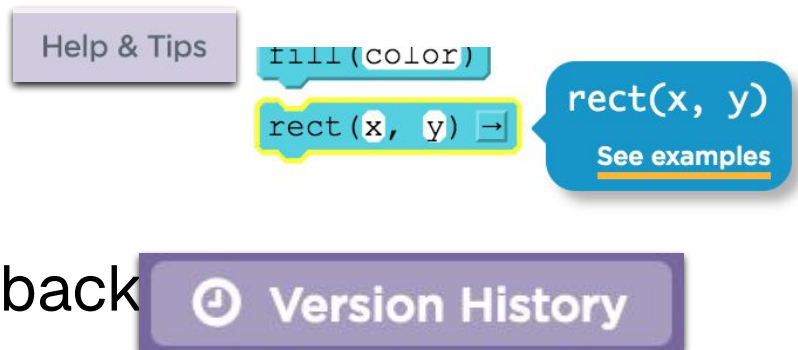
What are the steps to adding an image to a sprite?

Code Studio, Level 4-7

Skill Building

Do This

- Follow the instructions on each level
- If you get stuck, try...
 - Help and Tips Tab
 - Code Documentation
- If you make a big mistake...
 - Use version history to go back



Code Studio, Level 8

Match Level

Do This

- Match the code to the output

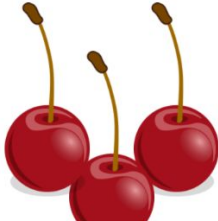
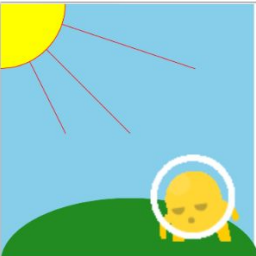
<pre>background(▼"skyblue"); drawSprites(); var dog = createSprite(100, 200); dog.setAnimation(▼"corgi"); var elk = createSprite(300, 200); elk.setAnimation(▼"elk");</pre>	?	
<pre>background(▼"skyblue"); var dog = createSprite(100, 200); drawSprites(); dog.setAnimation(▼"corgi"); var elk = createSprite(300, 200); elk.setAnimation(▼"elk");</pre>	?	
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Code Studio, Level 9

Practice Level

Do This

- Practice your new skills on these levels
- Check with your teacher before moving on

	a Where Are the Sprites? The sprites are being created, but you need to change the code to make the sprites appear.	 b Sprite Animations Fix the names of the animations so that the errors go away.	 c Sprite Name Fix the sprite name so that all three cherries appear.
	d Sprite Order Change the code so that we can see the ladybug and the snail.	 e Fish Scene Get creative and make a fish scene.	 f Combine Ideas Create your own scene using both drawing and sprites.



Code Studio, Level 10

Assessment Level

Do This

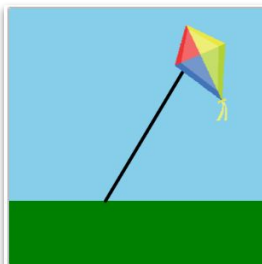
- Follow the instructions in Code Studio
- The "Rubric" tab describes how to demonstrate your understanding
- Check out the Help & Tips tab if you need help

Code Studio, Level 11

Challenge Level

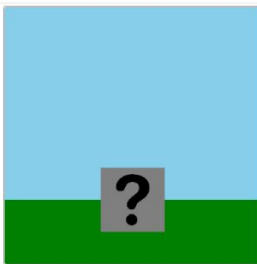
Do This

- Challenge yourself on these levels.
- Learn new blocks or use the same ones in new ways.



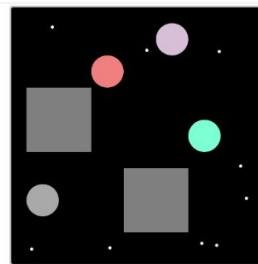
a Find an image online

Upload an image from another website to complete a kite scene.



b Draw Your Own Character

Create a sprite animation from scratch.



c Add Your Own Characters

Draw two sprites from scratch and add them to the scene.



d Fun Hairstyles

For this free play challenge, either draw your own sprites or edit existing sprites to give them your favorite hairstyles!



e Your Fashion Style

For this free play challenge, either draw your own sprites or edit existing sprites to give them your unique fashion style!

Wrap Up





Journal:

So far we've been able to change a sprite's location and image. What else might you want to change about your sprites?



Key Vocabulary

- **Sprite** - A character on the screen with properties that describe its location, movement, and look.



Question of the Day

How can we use sprites to help us keep track of lots of information in our programs?